

WEEK 4 REPORT – POWER BI EDA

Name: Maheswari C

Roll No: E25DA049

Department: B.Sc Data Science & AI

1. INTRODUCTION

This week, the **Shopify stock market data** was analyzed using **Microsoft Power BI**. The data was prepared for analysis, and DAX measures were created to study stock prices and trading volume.

Moving averages such as **20-Day SMA and 50-Day SMA** were used to understand the stock trend. Interactive charts, slicers, and a custom tooltip were also added to make the dashboard easier to use and understand.

2. DATA PREPARATION

The Shopify stock dataset was imported into Power BI and prepared for analysis.

The **Date** field was used to organize the stock data according to different time periods. The data was arranged to analyze stock prices and trading volume using Power BI visuals and DAX measures.

3. DAX MEASURES

DAX measures were created to calculate:

- 20-Day Simple Moving Average
- 50-Day Simple Moving Average
- Moving Average Crossover Signal

These measures were used to analyze moving-average trends in the Shopify stock data.

to select the required date period and `AVERAGE()` is used to calculate the average.

DAX Measure

```
Volume 20 SMA =  
VAR CurrentDate = MAX('Date Table'[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod =
```

```

        DATESINPERIOD(
            'Date Table'[Date],
            CurrentDate,
            -PeriodDays,
            DAY
        )
    )
RETURN
CALCULATE(
    AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),
    DateInPeriod
)

```

50-Day Simple Moving Average

A **50-Day Simple Moving Average (50 SMA)** was created to provide a longer-term comparison with the 20-day average.

DAX Measure

```

Volume 50 SMA =
VAR CurrentDate = MAX('Date Table'[Date])
VAR PeriodDays = 50
VAR DateInPeriod =
    DATESINPERIOD(
        'Date Table'[Date],
        CurrentDate,
        -PeriodDays,
        DAY
    )
RETURN
CALCULATE(
    AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),
    DateInPeriod
)

```

Moving Average Crossover Signal

A Moving Average Crossover Signal was created by comparing the **20 SMA** and **50 SMA**.

```

MA Crossover Signal =
IF(
    ISBLANK([Volume 20 SMA]) ||
    ISBLANK([Volume 50 SMA]),
    BLANK(),
    IF(
        [Volume 20 SMA] > [Volume 50 SMA],
        "Bullish (20 > 50)",
        "Bearish (20 < 50)"
    )
)

```

This measure follows the same crossover logic used in the reference report.

4. TRADING VOLUME ANALYSIS

The trading volume was analyzed to understand how stock activity changed over time.

A **20-Day Simple Moving Average (SMA)** and **50-Day Simple Moving Average** were used to understand the overall volume trend.

The chart shows changes in trading volume and helps identify periods of **high and low trading activity**. This makes it easier to understand changes in market activity during different time periods.

5. PRICE TREND ANALYSIS

The **Close Price** was analyzed to understand how the Shopify stock price changed over time.

The price chart shows the movement of the stock from **2015 to 2025**, including rises and falls in prices during different periods.

The **Daily Return %** was also used to observe daily changes in the stock price. This helps in understanding the overall price movement and changes in stock performance.

5.1 High and Low Price Trend

A line chart was created to compare the **High Price and Low Price** of Shopify stock over different dates.

Visual Elements

X-Axis: Date

Values: High Price, Low Price

Chart: Line Chart

This helps in understanding the range of stock price movement.

5.2 Open and Close Price Trend

Another line chart was created to compare the **Open Price and Close Price**.

Visual Elements

X-Axis: Date

Values: Open Price, Close Price

Chart: Line Chart

6. SLICERS AND DYNAMIC SELECTION

A **Month slicer** was added to the dashboard to filter the stock data based on different months.

This allows the user to select the required time period easily. The slicer makes the dashboard more interactive and helps users focus on the stock price and trading volume data they want to analyze.

7. CUSTOM TOOLTIP

A custom **Stock_Tooltip** page was created to show important stock details when the user moves the mouse over the chart.

The tooltip displays:

- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume
- Daily Change %

This makes it easier to view detailed stock information without adding too much data to the main dashboard.

8. CONCLUSION

The Week 4 analysis improved the **Shopify stock market dashboard** by using DAX measures and moving averages to study the stock data.

Trading volume and price trends were analyzed to understand changes over time. A **Month slicer** and interactive dashboard features were added to make the analysis easier.

The custom tooltip provided additional stock details such as price and volume while viewing the charts.

Overall, these features made the Power BI dashboard more **interactive and easier to understand**.